VF Asia Footwear Laboratory Testing Manual

Test Method for Rebound Resiliency	**Test method # VF- B16** **REF # : TM-R-028** **Page 1 of 2**
Date Issued : April , 2011 **Revision # :**	**Created by : Nila Babia Erazo** **Chris Adam**

PURPOSE AND SCOPE: To measure the impact resiliency by dropping a mass onto the surface of the test material and measuring the % rebound resiliency.

REFERENCES: DIN 53512-2000, DIN 53573 shapes A, DIN 53573 shapes B, ISO4662

EQUIPMENT AND MATERIAL:

1. Resiliometer, meeting the requirement of DIN 53512-2000 , DIN 53573 and ISO 466
2. Cellular , foam material
3. Non- cellular, elastic material, EVA , PU, PVC and rubber

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory conditions at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.
2. Select flat specimen, and prepare 2 to 3 specimens from different positions of the same batch.
3. Die or hand cut material with the following dimensions:

   - For non- cellular or elastic material , EVA , PU , PVC and Rubber : cut material with diameter of 50 mm and 12.5 mm in thickness
   - For cellular, foam material: cut material with dimensions of 80 mm in length, 80 mm in width and 50 mm in thickness.

REPETITIONS: Test three (3) samples per material. (i.e., nine total readings – three readings from each of three samples.

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.

1. Check the following at the start of each test :
   - The machine is level , if not , level it by using the footpads together with the spirit level
   - The pendulum is fixed in the released handle.

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2. Screw off the fixed nut and attach the striker head to the pendulum with the head downward vertically. Ensure that the attached striker head is correct:
    - For non- cellular or elastic material, EVA, PU, PVC and rubber: use the impact energy of 0.5 J . Ball diameter of 15mm which comply with DIN 53512 & ISO 4662.
    - For cellular, foam material: use the impact energy of 0.2 J and Shape B cylinder diameter - 40mm , curvature radius of the round end at 40mm and which comply with DIN 53573
3. Clamp the specimens according to the type of material:
    - For non- cellular or elastic material , EVA , PU , PVC and Rubber :
        - Place the specimen on the two round pins of the circular station.
        - Push the clamping rod to make the arc clips, hold the specimen.
    - For cellular , foam material :
        - Place the specimen in the recess of the square station (note: remove the circular station before operating)
        - Disuse the arc clips, meaning it should be moved downward so that the striker head will not hit the arc clips.
4. Proceed with the test using the following steps:
    - Place the pendulum in a vertical position then start the release handle. Adjust the span between the pendulum and specimen with the adjuster so that they are in contact slightly, and then lock the moving axis by the fixed knob.
    - Again fix the pendulum into the release handle ; turn the turn knob to make the pointer point to 0%
    - Start the release handle to allow the pendulum to hit the same position of the specimen three times. Ensure that the surface is flat or level and the pointer is pointing at 0% before starting each hit.
    - Test six times for the same position, the first three hits should not be recorded; this is for adjusting the mechanical performance of the specimen.

REPORT: The average of the nine readings from the three test samples to the nearest 0.1%.